

$$\frac{(k+1)^2/2^{k+1}}{k^2/2^k} = \frac{(k+1)^2}{2k^2} \leq 8/9$$

if $k \geq 3$. Thus, you can split the summation into

$$\begin{aligned} \sum_{k=0}^{\infty} \frac{k^2}{2^k} &= \sum_{k=0}^2 \frac{k^2}{2^k} + \sum_{k=3}^{\infty} \frac{k^2}{2^k} \\ &= \sum_{k=0}^2 \frac{k^2}{2^k} + \sum_{k=0}^{\infty} \frac{(k+3)^2}{2^{k+3}} \quad (\text{by reindexing}) \\ &\leq \sum_{k=0}^2 \frac{k^2}{2^k} + \frac{9}{8} \sum_{k=0}^{\infty} \left(\frac{8}{9}\right)^k \quad (\text{by inequality (A.15)}) \\ &= (0 + 1/2 + 1) + \frac{9/8}{1 - 8/9} \quad (\text{by equation (A.16)}) \\ &= O(1). \end{aligned}$$

The technique of splitting summations can help determine asymptotic bounds in much more difficult situations. For example, here is one way to obtain a bound of $O(\lg n)$ on the harmonic series (A.9):

$$H_n = \sum_{k=1}^n \frac{1}{k}.$$

The idea is to split the range 1 to n into $\lfloor \lg n \rfloor + 1$ pieces and upper-bound the contribution of each piece by 1. For $i = 0, 1, \dots, \lfloor \lg n \rfloor$, the i th piece consists of the terms starting at $1/2^i$ and going up to but not including $1/2^{i+1}$. The last piece might contain terms not in the original harmonic series, giving

$$\begin{aligned} \sum_{k=1}^n \frac{1}{k} &\leq \sum_{i=0}^{\lfloor \lg n \rfloor} \sum_{j=0}^{2^i-1} \frac{1}{2^i + j} \\ &\leq \sum_{i=0}^{\lfloor \lg n \rfloor} \sum_{j=0}^{2^i-1} \frac{1}{2^i} \\ &= \sum_{i=0}^{\lfloor \lg n \rfloor} \left(2^i \cdot \frac{1}{2^i}\right) \\ &= \sum_{i=0}^{\lfloor \lg n \rfloor} 1 \\ &\leq \lg n + 1. \end{aligned} \tag{A.17}$$

Approximation by integrals

When a summation has the form $\sum_{k=m}^n f(k)$, where $f(k)$ is a monotonically increasing function, you can approximate it by integrals:

$$\int_{m-1}^n f(x) dx \leq \sum_{k=m}^n f(k) \leq \int_m^{n+1} f(x) dx . \quad (\text{A.18})$$

Figure A.1 justifies this approximation. The summation is represented as the area of the rectangles in the figure, and the integral is the blue region under the curve. When $f(k)$ is a monotonically decreasing function, you can use a similar method to provide the bounds

$$\int_m^{n+1} f(x) dx \leq \sum_{k=m}^n f(k) \leq \int_{m-1}^n f(x) dx . \quad (\text{A.19})$$

The integral approximation (A.19) can be used to prove the tight bounds in inequality (A.10) for the n th harmonic number. The lower bound is

$$\begin{aligned} \sum_{k=1}^n \frac{1}{k} &\geq \int_1^{n+1} \frac{dx}{x} \\ &= \ln(n+1) , \end{aligned} \quad (\text{A.20})$$

For the upper bound, the integral approximation gives

$$\begin{aligned} \sum_{k=1}^n \frac{1}{k} &= \sum_{k=2}^n \frac{1}{k} + 1 \\ &\leq \int_1^n \frac{dx}{x} + 1 \\ &= \ln n + 1 . \end{aligned} \quad (\text{A.21})$$

Exercises

A.2-1

Show that $\sum_{k=1}^n 1/k^2$ is bounded above by a constant.

A.2-2

Find an asymptotic upper bound on the summation

$$\sum_{k=0}^{\lfloor \lg n \rfloor} \lceil n/2^k \rceil .$$

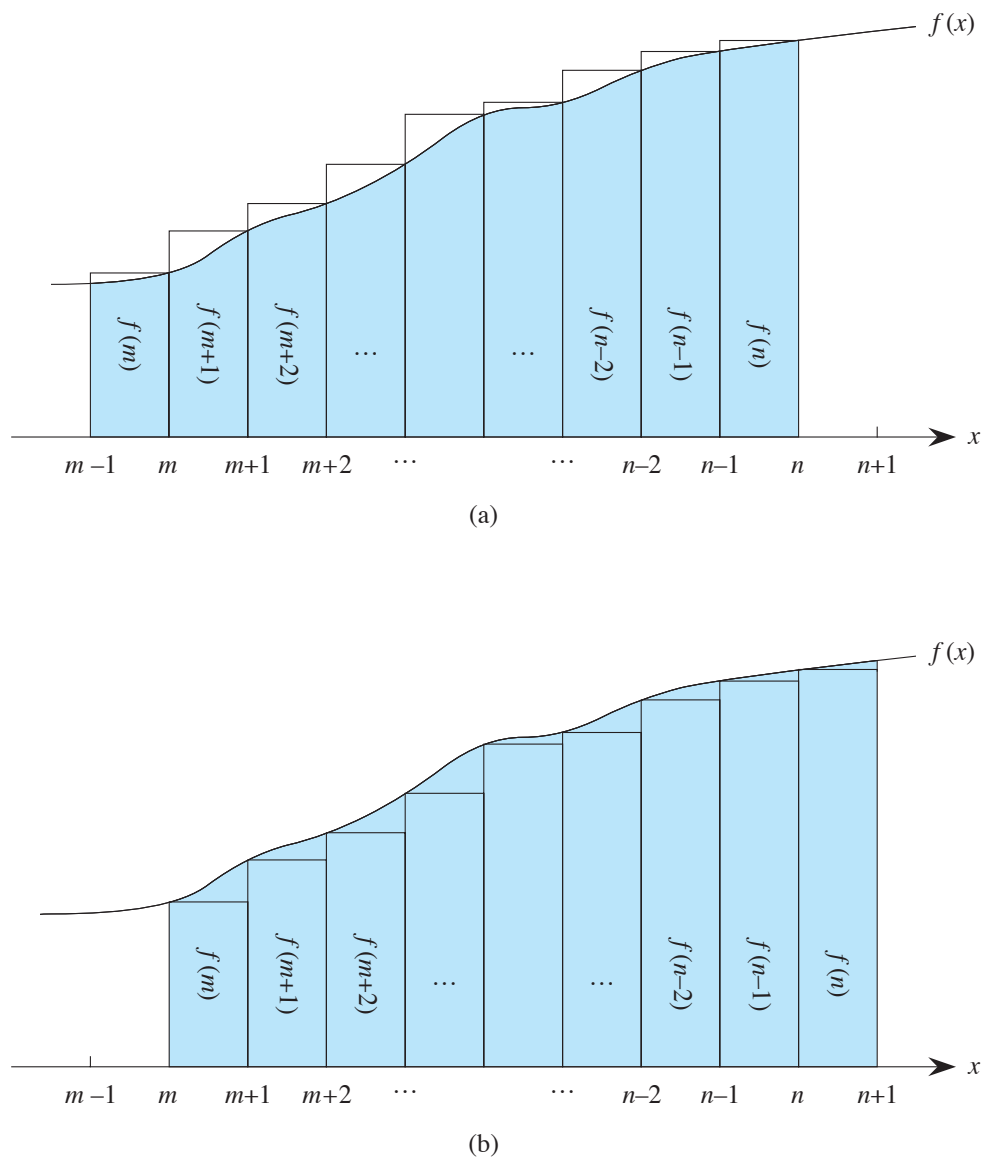


Figure A.1 Approximation of $\sum_{k=m}^n f(k)$ by integrals. The area of each rectangle is shown within the rectangle, and the total rectangle area represents the value of the summation. The integral is represented by the blue area under the curve. Comparing areas in **(a)** gives the lower bound $\int_{m-1}^n f(x) dx \leq \sum_{k=m}^n f(k)$. Shifting the rectangles one unit to the right gives the upper bound $\sum_{k=m}^n f(k) \leq \int_m^{n+1} f(x) dx$ in **(b)**.

A.2-3

Show that the n th harmonic number is $\Omega(\lg n)$ by splitting the summation.

A.2-4

Approximate $\sum_{k=1}^n k^3$ with an integral.

A.2-5

Why can't you use the integral approximation (A.19) directly on $\sum_{k=1}^n 1/k$ to obtain an upper bound on the n th harmonic number?

Problems
A-1 Bounding summations

Give asymptotically tight bounds on the following summations. Assume that $r \geq 0$ and $s \geq 0$ are constants.

a. $\sum_{k=1}^n k^r.$

b. $\sum_{k=1}^n \lg^s k.$

c. $\sum_{k=1}^n k^r \lg^s k.$

Appendix notes

Knuth [259] provides an excellent reference for the material presented here. You can find basic properties of series in any good calculus book, such as Apostol [19] or Thomas et al. [433].

B Sets, Etc.

Many chapters of this book touch on the elements of discrete mathematics. This appendix reviews the notations, definitions, and elementary properties of sets, relations, functions, graphs, and trees. If you are already well versed in this material, you can probably just skim this chapter.

B.1 Sets

A *set* is a collection of distinguishable objects, called its *members* or *elements*. If an object x is a member of a set S , we write $x \in S$ (read “ x is a member of S ” or, more briefly, “ x belongs to S ”). If x is not a member of S , we write $x \notin S$. To describe a set explicitly, write its members as a list inside braces. For example, to define a set S to contain precisely the numbers 1, 2, and 3, write $S = \{1, 2, 3\}$. Since 2 belongs to the set S , we can write $2 \in S$, and since 4 is not a member, we can write $4 \notin S$. A set cannot contain the same object more than once,¹ and its elements are not ordered. Two sets A and B are *equal*, written $A = B$, if they contain the same elements. For example, $\{1, 2, 3, 1\} = \{1, 2, 3\} = \{3, 2, 1\}$.

We adopt special notations for frequently encountered sets:

- $\emptyset$ denotes the *empty set*, that is, the set containing no members.
- $\mathbb{Z}$ denotes the set of *integers*, that is, the set $\{\dots, -2, -1, 0, 1, 2, \dots\}$.
- $\mathbb{R}$ denotes the set of *real numbers*.
- $\mathbb{N}$ denotes the set of *natural numbers*, that is, the set $\{0, 1, 2, \dots\}$.²

¹ A variation of a set, which can contain the same object more than once, is called a *multiset*.

² Some authors start the natural numbers with 1 instead of 0. The modern trend seems to be to start with 0.

If all the elements of a set A are contained in a set B , that is, if $x \in A$ implies $x \in B$, then we write $A \subseteq B$ and say that A is a **subset** of B . A set A is a **proper subset** of set B , written $A \subset B$, if $A \subseteq B$ but $A \neq B$. (Some authors use the symbol “ $\subset$ ” to denote the ordinary subset relation, rather than the proper-subset relation.) Every set is a subset of itself: $A \subseteq A$ for any set A . For two sets A and B , we have $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$. The subset relation is transitive (see page 1159): for any three sets A , B , and C , if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$. The proper-subset relation is transitive as well. The empty set is a subset of all sets: for any set A , we have $\emptyset \subseteq A$.

Sets can be specified in terms of other sets. Given a set A , a set $B \subseteq A$ can be defined by stating a property that distinguishes the elements of B . For example, one way to define the set of even integers is $\{x : x \in \mathbb{Z} \text{ and } x/2 \text{ is an integer}\}$. The colon in this notation is read “such that.” (Some authors use a vertical bar in place of the colon.)

Given two sets A and B , **set operations** define new sets:

- The **intersection** of sets A and B is the set

$$A \cap B = \{x : x \in A \text{ and } x \in B\} .$$

- The **union** of sets A and B is the set

$$A \cup B = \{x : x \in A \text{ or } x \in B\} .$$

- The **difference** between two sets A and B is the set

$$A - B = \{x : x \in A \text{ and } x \notin B\} .$$

Set operations obey the following laws:

Empty set laws:

$$A \cap \emptyset = \emptyset ,$$

$$A \cup \emptyset = A .$$

Idempotency laws:

$$A \cap A = A ,$$

$$A \cup A = A .$$

Commutative laws:

$$A \cap B = B \cap A ,$$

$$A \cup B = B \cup A .$$

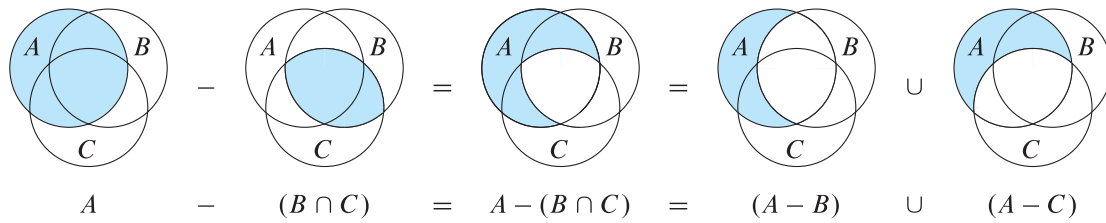


Figure B.1 A Venn diagram illustrating the first of DeMorgan's laws (B.2). Each of the sets A , B , and C is represented as a circle.

Associative laws:

$$\begin{aligned} A \cap (B \cap C) &= (A \cap B) \cap C, \\ A \cup (B \cup C) &= (A \cup B) \cup C. \end{aligned}$$

Distributive laws:

$$\begin{aligned} A \cap (B \cup C) &= (A \cap B) \cup (A \cap C), \\ A \cup (B \cap C) &= (A \cup B) \cap (A \cup C). \end{aligned} \tag{B.1}$$

Absorption laws:

$$\begin{aligned} A \cap (A \cup B) &= A, \\ A \cup (A \cap B) &= A. \end{aligned}$$

DeMorgan's laws:

$$\begin{aligned} A - (B \cap C) &= (A - B) \cup (A - C), \\ A - (B \cup C) &= (A - B) \cap (A - C). \end{aligned} \tag{B.2}$$

Figure B.1 illustrates the first of DeMorgan's laws, using a *Venn diagram*: a graphical picture in which sets are represented as regions of the plane.

Often, all the sets under consideration are subsets of some larger set U called the *universe*. For example, when considering various sets made up only of integers, the set $\mathbb{Z}$ of integers is an appropriate universe. Given a universe U , we define the *complement* of a set A as $\overline{A} = U - A = \{x : x \in U \text{ and } x \notin A\}$. For any set $A \subseteq U$, we have the following laws:

$$\begin{aligned} \overline{\overline{A}} &= A, \\ A \cap \overline{A} &= \emptyset, \\ A \cup \overline{A} &= U. \end{aligned}$$

An equivalent way to express DeMorgan's laws (B.2) uses set complements. For any two sets $B, C \subseteq U$, we have

$$\overline{B \cap C} = \overline{B} \cup \overline{C},$$

$$\overline{B \cup C} = \overline{B} \cap \overline{C}.$$

Two sets A and B are *disjoint* if they have no elements in common, that is, if $A \cap B = \emptyset$. A *collection* of sets $S_1, S_2, \dots$, either finite or infinite, is a set of sets, in which each member is a set S_i . A collection $\mathcal{S} = \{S_i\}$ of nonempty sets forms a *partition* of a set S if

- the sets are *pairwise disjoint*, that is, $S_i, S_j \in \mathcal{S}$ and $i \neq j$ imply $S_i \cap S_j = \emptyset$, and
- their union is S , that is,

$$S = \bigcup_{S_i \in \mathcal{S}} S_i.$$

In other words, $\mathcal{S}$ forms a partition of S if each element of S appears in exactly one set $S_i \in \mathcal{S}$.

The number of elements in a set is the *cardinality* (or *size*) of the set, denoted $|S|$. Two sets have the same cardinality if their elements can be put into a one-to-one correspondence. The cardinality of the empty set is $|\emptyset| = 0$. If the cardinality of a set is a natural number, the set is *finite*, and otherwise, it is *infinite*. An infinite set that can be put into a one-to-one correspondence with the natural numbers $\mathbb{N}$ is *countably infinite*, and otherwise, it is *uncountable*. For example, the integers $\mathbb{Z}$ are countable, but the reals $\mathbb{R}$ are uncountable.

For any two finite sets A and B , we have the identity

$$|A \cup B| = |A| + |B| - |A \cap B|, \quad (\text{B.3})$$

from which we can conclude that

$$|A \cup B| \leq |A| + |B|.$$

If A and B are disjoint, then $|A \cap B| = 0$ and thus $|A \cup B| = |A| + |B|$. If $A \subseteq B$, then $|A| \leq |B|$.

A finite set of n elements is sometimes called an *n-set*. A 1-set is called a *singleton*. A subset of k elements of a set is sometimes called a *k-subset*.

We denote the set of all subsets of a set S , including the empty set and S itself, by 2^S , called the *power set* of S . For example, $2^{\{a,b\}} = \{\emptyset, \{a\}, \{b\}, \{a,b\}\}$. The power set of a finite set S has cardinality $2^{|S|}$ (see Exercise B.1-5).

We sometimes care about setlike structures in which the elements are ordered. An *ordered pair* of two elements a and b is denoted (a, b) and is defined formally

as the set $(a, b) = \{a, \{a, b\}\}$. Thus, the ordered pair (a, b) is *not* the same as the ordered pair (b, a) .

The **Cartesian product** of two sets A and B , denoted $A \times B$, is the set of all ordered pairs such that the first element of the pair is an element of A and the second is an element of B . More formally,

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\} .$$

For example, $\{a, b\} \times \{a, b, c\} = \{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c)\}$. When A and B are finite sets, the cardinality of their Cartesian product is

$$|A \times B| = |A| \cdot |B| . \tag{B.4}$$

The Cartesian product of n sets $A_1, A_2, \dots, A_n$ is the set of ***n*-tuples**

$$A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \dots, a_n) : a_i \in A_i \text{ for } i = 1, 2, \dots, n\} ,$$

whose cardinality is

$$|A_1 \times A_2 \times \cdots \times A_n| = |A_1| \cdot |A_2| \cdots |A_n|$$

if all sets A_i are finite. We denote an n -fold Cartesian product over a single set A by the set

$$A^n = \underbrace{A \times A \times \cdots \times A}_{n \text{ times}} ,$$

whose cardinality is $|A^n| = |A|^n$ if A is finite. We can also view an n -tuple as a finite sequence of length n (see page 1162).

Intervals are continuous sets of real numbers. We denote them with parentheses and/or brackets. Given real numbers a and b , the **closed interval** $[a, b]$ is the set $\{x \in \mathbb{R} : a \leq x \leq b\}$ of reals between a and b , including both a and b . (If $a > b$, this definition implies that $[a, b] = \emptyset$.) The **open interval** $(a, b) = \{x \in \mathbb{R} : a < x < b\}$ omits both of the endpoints from the set. There are two **half-open intervals** $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$ and $(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$, each of which excludes one endpoint.

Intervals can also be defined on the integers by replacing $\mathbb{R}$ in the these definitions by $\mathbb{Z}$. Whether the interval is defined over the reals or integers can usually be inferred from context.

Exercises

B.1-1

Draw Venn diagrams that illustrate the first of the distributive laws (B.1).

B.1-2

Prove the generalization of DeMorgan's laws to any finite collection of sets:

$$\overline{A_1 \cap A_2 \cap \cdots \cap A_n} = \overline{A_1} \cup \overline{A_2} \cup \cdots \cup \overline{A_n},$$

$$\overline{A_1 \cup A_2 \cup \cdots \cup A_n} = \overline{A_1} \cap \overline{A_2} \cap \cdots \cap \overline{A_n}.$$

★ B.1-3

Prove the generalization of equation (B.3), which is called the *principle of inclusion and exclusion*:

$$\begin{aligned} |A_1 \cup A_2 \cup \cdots \cup A_n| = & \\ & |A_1| + |A_2| + \cdots + |A_n| \\ & - |A_1 \cap A_2| - |A_1 \cap A_3| - \cdots \quad (\text{all pairs}) \\ & + |A_1 \cap A_2 \cap A_3| + \cdots \quad (\text{all triples}) \\ & \vdots \\ & + (-1)^{n-1} |A_1 \cap A_2 \cap \cdots \cap A_n|. \end{aligned}$$

B.1-4

Show that the set of odd natural numbers is countable.

B.1-5

Show that for any finite set S , the power set 2^S has $2^{|S|}$ elements (that is, there are $2^{|S|}$ distinct subsets of S).

B.1-6

Give an inductive definition for an n -tuple by extending the set-theoretic definition for an ordered pair.

B.2 Relations

A *binary relation* R on two sets A and B is a subset of the Cartesian product $A \times B$. If $(a, b) \in R$, we sometimes write $a R b$. When we say that R is a binary relation on a set A , we mean that R is a subset of $A \times A$. For example, the “less than” relation on the natural numbers is the set $\{(a, b) : a, b \in \mathbb{N} \text{ and } a < b\}$. An n -ary relation on sets $A_1, A_2, \dots, A_n$ is a subset of $A_1 \times A_2 \times \cdots \times A_n$.

A binary relation $R \subseteq A \times A$ is *reflexive* if

$$a R a$$